

Polygon training: image size & augmentation review

Task: TreePolygons (Mask R-CNN, DeepForest stack, COCO init, lr 0.01). **Data:** v0.20 unless noted. **Metrics:** Recall / Mask-aware Precision / AP50 on the test split. Comet project: [bw4sz/milliontrees-polygons](#). **Context:** ~75% of training images are only partially annotated; the prior torchvision stack reached AP50 ≈ 0.28 (within) — the DeepForest stack has not yet matched it.

1. Eval image size (fixed model, vary eval input only)

Same trained checkpoint; swept only the evaluation image_size. Mask R-CNN already rescales internally (~800 px), so larger input shrinks trees relative to anchors.

Eval image size	Recall	Mask-aware Prec	AP50
448 (≈ train scale)	0.604	0.858	0.284
672	0.578	0.859	0.281
896	0.535	0.861	0.249
1280	0.478	0.861	0.215

Takeaway: recall and AP50 fall monotonically as eval size grows; precision flat. Best at native train scale (448). Bigger ≠ better.

2. Train input strategy (within-distribution, v0.20)

Strategy	Train input	Eval mode	Recall	Prec	AP50	Comet
Resize 640 (control)	whole img → resize+pad 640	resize 640	0.382	0.882	0.165	ece8bf
Native crop 640	RandomCrop 640 @ native GSD	tiled native	0.371	0.816	0.083	639e2b
RandomResizedCrop	scale 0.64–1.0 → 640	tiled native	0.341	0.812	0.081	cbf4df
Resize 1280	whole img → 1280	resize 1280	0.004	0.004	coll. ¹	c0577d
Annotation-safe crop	RandomSizedBBBoxSafeCrop	tiled native — in progress —				40a118

¹ Within-distribution 1280 training diverged (val_loss → NaN). OOD 1280 survived but trailed 640 (below).

Takeaway: whole-image resize to 640 with scale-matched eval wins (AP50 0.165, prec 0.88). Native/random crops lose ~2× AP50 — crops land on unlabeled trees in partially-annotated images, injecting false negatives. 1280 is too aggressive (trees ≈ 4 px, training destabilizes). Annotation-safe crop is the in-flight fix.

3. Out-of-distribution split

Strategy	Recall	Mask-aware Prec	AP50	Comet
Resize 640	0.332	0.843	0.110	193d60
Resize 1280	0.217	0.782	0.076	588471

Same ranking as within-distribution: 640 > 1280.

4. Eval projection: squish-448 → tiled native-res (DeepForest-integration fix)

Eval path	Recall	Mask-aware Prec	AP50
Squish-448 (old)	0.244	0.662	0.032
Tiled native-res (fix)	0.249	0.668	0.052

Bottom line

- Use **resize-640 + scale-matched eval** as the polygon recipe — best AP50/precision, stable.
- **Don't push image size up** (1280 hurts or breaks; eval-size sweep peaks at 448).
- **Crops underperform** due to incomplete annotation — testing **annotation-safe crop** next.
- DeepForest stack at AP50 0.165 still trails the old torchvision 0.28; closing that gap is the open item.

Image comparisons

Train input strategy — same scene (Ball et al. 2023, Paracou; within-distribution, v0.20). **Orange** = model prediction, **purple** = ground truth. Resize-640 and the crops actively predict crowns; **resize-1280 predicts almost nothing** (recall 0.004).



Resize 640 — best (AP50 0.165)



Native crop 640 (AP50 0.083)



RandomResizedCrop (AP50 0.081)



Resize 1280 — collapsed (recall 0.004)